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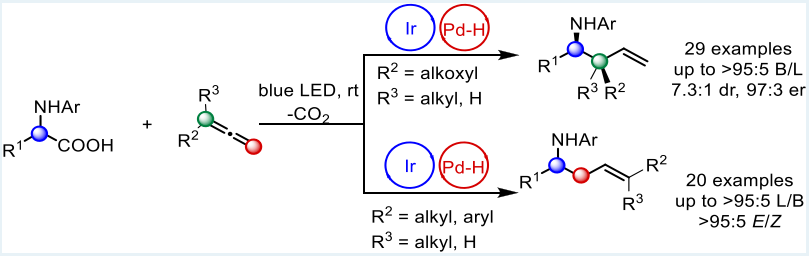
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Dual Palladium/Photoredox-Catalyzed Enantioselective and Regioselective Decarboxylative Hydroaminoalkylation of Allenes

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ABSTRACT: A dual palladium/photoredox-catalyzed regio- and enantioselective decarboxylative hydroaminoalkylation of allenes with amino acids is reported. This mild, redox-neutral, and atom-economic process allows access to a range of protected vinyl 1,2-amino ethers in good yields with high levels of branched and enantioselectivity, when alkoxyallenes were employed. Conversely, with alkyl- and aryl-substituted terminal allenes as the substrates, a linear-selective decarboxylative hydroaminoalkylation of allenes is realized in high regio- and *E*/*Z* selectivities with a Pd/*rac*-BINAP catalytic system.

KEYWORDS: *α-amino acid, allene, palladium, photoredox, chiral vicinal amino alcohol*

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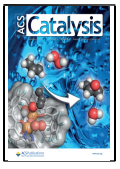
Chiral 1,2-amino alcohols have emerged as a common motif in a variety of different natural products and pharmaceutically active compounds as well as privileged chiral auxiliaries and ligands (Figure 1). As a result, the importance of vicinal amino alcohols has stimulated extensive effort from the synthetic community toward their synthesis.[1] Among them, processes that form the C−C bond between the amine and alcohol functionalities are particularly attractive but challenging using typical two-electron ionic processes (Scheme 1A).

Therefore, a traditional reverse polarity (Umpolung) strategy, such as using nitroalkane nucleophiles as carbanion equivalents, has been proposed for addressing this issue.[2] Despite their effectiveness, these methods often require subsequent refunctionalization to deliver the amine. The direct enantioselective *α*-lithiation of alkylamines for the addition to carbonyls is an alternative, which nevertheless often requires cryogenic conditions, the use of stoichiometric amounts of sparteine and strong bases (alkyl lithiums).[3] Hence, developing a streamlined, efficient, and atom-economic methodology to enable the enantioselective construction of vicinal amino alcohols via a C−C bond formation process under ambient conditions is highly demanded.

In recent years, radical cross-coupling between amine and alcohol moieties for the construction of vicinal amino alcohols by photoredox catalysis has received widespread attention.[4] However, only a few examples of asymmetric photoredox catalysis to access enantioenriched this class of compounds have been reported,[5] including chiral phosphoric acid organocatalysts,[6] chiral-at-metal rhodium/iridium complex,[7] and chiral scandium-*N*,*N*′-dioxide complex.[8] Furthermore, long-living ketyl radicals obtained from a reduction of aromatic carbonyls are required for a successful radical cross-coupling in all of these methods, which diminish the generality of the scope. Therefore, exploring alternative alcohol moieties that can enable versatile further transformations and effectively couple with *α*-amino alkyl radicals is highly desirable.

Alkoxyallenes are versatile building blocks for organic synthesis, especially serving as atom-economic[9] *π*-allyl fragment precursors for palladium-catalyzed allylic addition to pronucleophiles leading to allylic alcohol building blocks.[10] However, the current state of the art to access C−C bonds is still restricted to the allylation with soft carbon nucleophiles, such as Meldrum's acids,[11] azlactones,[12] oxindoles,[13] and pyrazolones[14] (Scheme 1B). To broaden the carbon pronucleophiles window of hydroalkylation of allenes, we recently described a rhodium/photoredox co-catalyzed atom-economic regiodivergent hydroaminoalkylation of alkynes and allenes, but unfortunately only racemic homoallylic amines could be obtained.[15]

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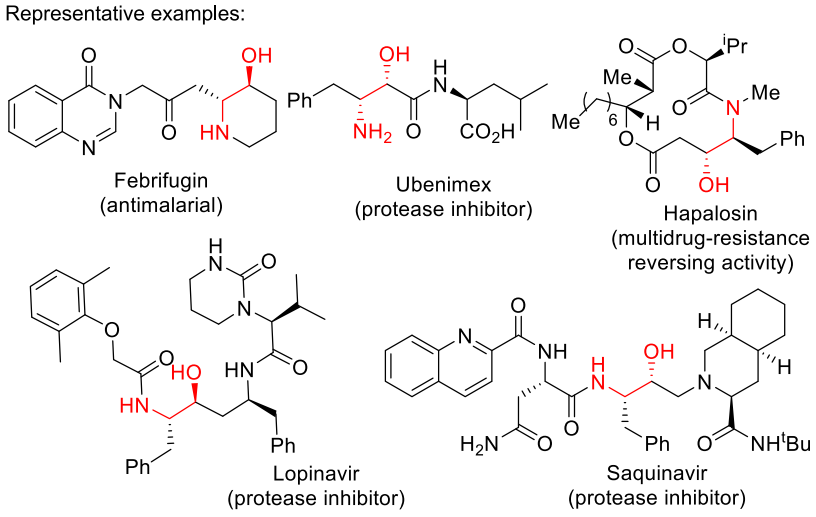
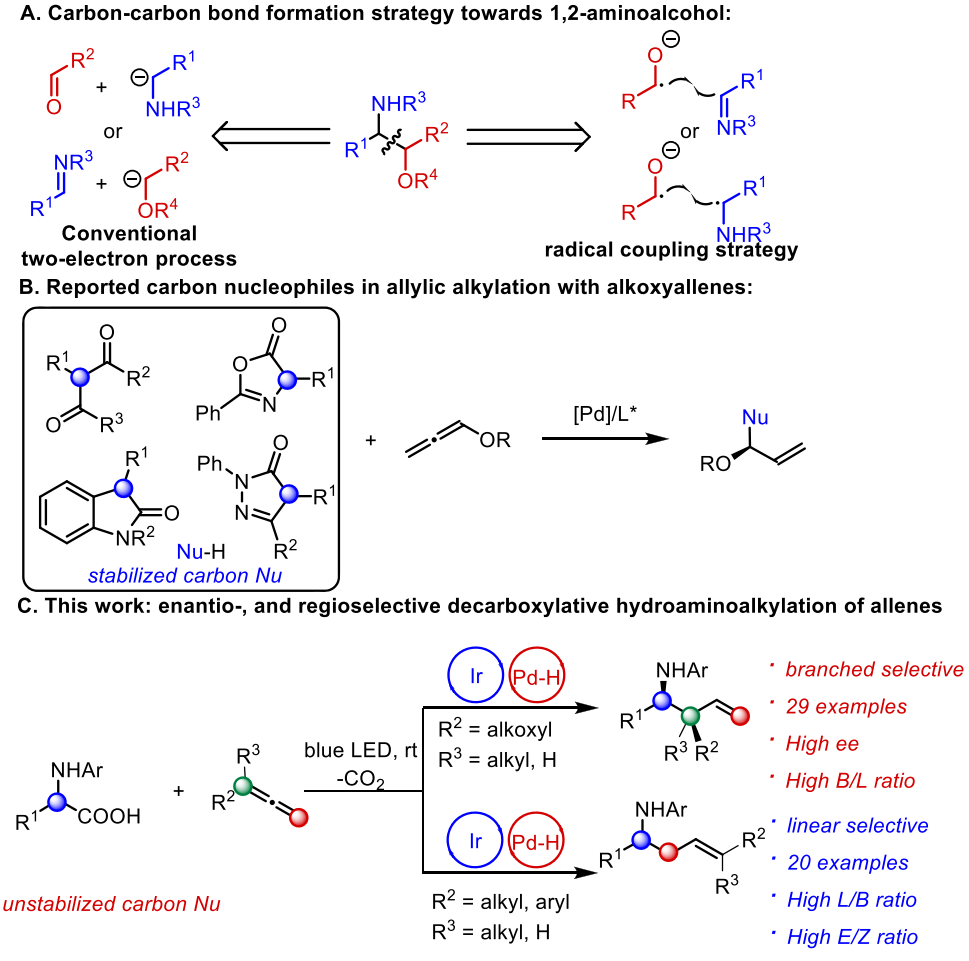


Figure 1. Representative examples on 1,2-amino alcohol-containing bioactive compounds.

Scheme 1. (A) Retrosynthetic Analysis of 1,2-Amino Alcohols. (B) Reported Soft Carbon Nucleophiles Used in Allylic Alkylation of Alkoxyallenes. (C) This Work: Regioselective Decarboxylative Hydroaminoalkylation of Allenes for the Synthesis of Enantioenriched Protected 1,2-Amino Alcohols



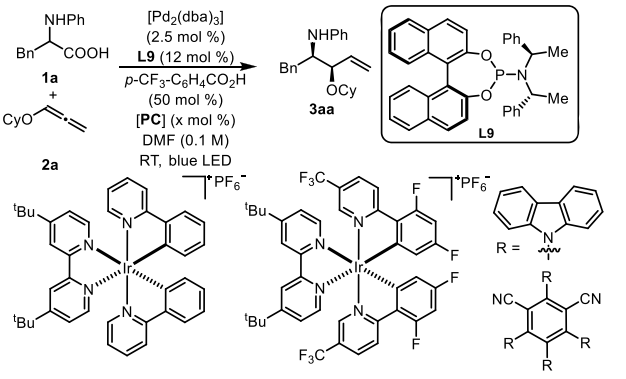
Inspired by these advances as well as recent progress in enantioselective metallaphotoredox catalysis,[5] we envisioned that the utility of alkoxyallenes as the allylic precursor in the cross-coupling with α-amino alkyl radicals could lead to the formation of 1,2-amino ethers (Scheme 1C). In addition, easy-to-access α-amino acids, which are widely used to generate α-amino alkyl radicals via a single-electron-oxidative decarboxylation process,[16] could be suitable substrates for the hydroaminoalkylation of allenes because α-amino acids can simultaneously serve as reductants and a proton source. The released proton from the decarboxylation of α-amino acids could promote the generation of a palladium(II) hydride

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species, which is crucial for the formation of a π-allyl palladium intermediate from allenes. Moreover, the π-allyl palladium species is believed to be the key intermediate to control reactivity, regioselectivity, and stereoselectivity.[17] After being attacked by the activated α-amino alkyl radical and one electron reduction, the desired 1,2-amino alcohol function might be obtained. Herein, we report on an enantioselective decarboxylative hydroaminoalkylation of allenes with α-amino acids via a combined palladium and photoredox catalysis to give facile access to *syn*-vicinal amino alcohols with high levels of regio- and enantiocontrol.

We began our studies by reacting *N*-phenyl-protected phenyl alanine (**1a**) and cyclohexyloxyallene (**2a**) in the presence of 2.5 mol % of $[Pd_2(dba)_3]$, 6 mol % (*S*)-BINAP, 50 mol % *p*-$CF_3C_6H_4COOH$, 1 mol % $[Ir(ppy)_2(dtbbpy)PF_6]$ [**PC1**], and DMF as solvents under bluelight-emitting-diode (LED) irradiation. After 16 h, the desired product **3aa** was obtained in good yields, with >19:1 B/L but poor diastereo- and enantiocontrol (1.3:1 dr, 69:31 er).[18] To our delight, further ligand screening revealed that the Feringa ligand (**L9**) proved to be optimal, affording product **3aa** with 72% yield, 86:14 B/L, 4:1 dr, and 97:3 er (Table 1, entry 1).[18] A survey of solvents and acid additives was conducted, however, without beneficial results.[18] Interestingly, evaluation of different photosensitizers resulted in a slight improvement in regioselectivity (90:10 B/L) using $[Ir(dF(CF_3)ppy)_2(dtbbpy)PF_6]$ [**PC2**] as the photocatalyst, while maintaining reactivity (70% yield) and stereoselectivity (4.5:1 dr, 96:4 er) (Table 1, entries 2 and 3). Variation of the photocatalyst loading revealed that 1 mol % provided optimal results (Table 1, entries 4−5). Finally, control experiments demonstrated that photocatalyst, palladium catalyst, and blue LED are all crucial for this decarboxylative α-allylation reaction (Table 1, entries 6−8).

Table 1. Optimization of Reaction Conditions[a]



entry	PC	x	yield/%	B/L	dr	er
1	**PC1**	1	72	86:14	4:1	97:3/ 89:11
2	***PC2***	***1***	***73(70)***	***90:10***	***4.5:1***	***96:4/88:12***
3	**PC3**	1	41	85:15	3.2:1	96:4/88:12
4	**PC2**	2	67	89:11	4.5:1	96:4/90:10
5	**PC2**	0.5	56	90:10	4.2:1	96:4/88:12
6[b]			nr			
7[c]	**PC2**	1	nr			
8[d]	**PC2**	1	nr			

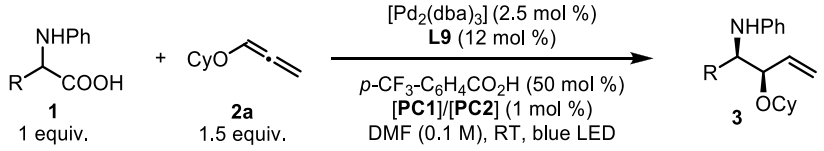
[a]Reaction conditions (unless otherwise specified): **1a** (0.1 mmol), **2a** (0.15 mmol), $[Pd_2(dba)_3]$ (2.5 mol %), **L9** (12 mol %), *p*-$CF_3C_6H_4COOH$ (50 mol %), [**PC**] (*x* mol %), DMF (1 mL), 16 h, room temperature (rt), blue LED. NMR yields are reported by using isoquinoline as internal standard, and isolated yield is presented in parentheses. Diastereomeric ratio (dr) was determined by ^{1}H NMR spectroscopy of the crude mixture. Enantiomeric ratio (er) was determined by chiral high-performance liquid chromatography (HPLC) analysis. [b]Without [**PC**]. [c]Without blue LED. [d]Without Pd and ligand.

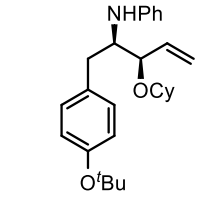
With the optimized reaction conditions in hand, the substrate scope with respect to the two reaction partners was investigated. First, we explored *N*-phenyl-protected amino acids derived from natural and unnatural amino acids. As summarized in Table 2, a wide range of branched allylated 1,2-amino ethers can readily be obtained in good yields with high regio- and enantioselectivities. In addition to *O*-*tert*-butyl-tyrosine, unprotected tyrosine was also found to be a suitable substrate yielding the allylation adduct (**3ca**) with 91:9 regioselectivity and 96:4 enantiomeric ratio, as well as complete chemoselectivity in favor of carbon−carbon bond formation instead of the phenol acting as an oxygen nucleophile. A 2-naphthyl alanine derivative was allylated well in this Pd/Ir dual catalytic system, affording the corresponding product **3da** in 74% yield with 96:4 er. Furthermore, a tryptophan derivative with an unprotected indole moiety coupled well to furnish allylation adduct **3ea** in good yield and high control of regio- and enantioselectivities. Other *N*-phenyl-derived amino acids with aliphatic side chains, including alanine, 2-aminobutanoic acid, norvaline, and leucine, were also successfully allylated, yielding the related 1,2-amino ethers (**3fa−3ja**) in 41−55% yields with high regio- and enantioselectivity. In addition, *N*-phenyl amino acids bearing other functional groups such as OBn, O^{t}Bu, SMe, and carboxylate esters, involving serine, methionine, aspartic acid, and glutamic acid, were also compatible in this hydroaminoalkylation reaction. In all cases, the products (**3ka−3oa**) were obtained in moderate diastereoselectivity with high regio- and enantiocontrol. To be noted, a cyclic amino acid derived from proline coupled well with the allene, resulting in a tertiary homoallylic amine (**3pa**) in 93:7 er.

As demonstrated in Table 3, with *N*-phenyl-protected phenyl alanine (**1a**) as the coupling partner under the standard reaction conditions, the transformation worked well for a range of alkoxyallenes. The enantioselective hydroaminoalkylation of cyclopentyl, cycloheptyl, (trimethylsilyl)ethyl, and nonyl-oxyallenes also proceeded smoothly to furnish the corresponding allylated adducts (**3ab−3ae**) in good yields (71−78% yields) and diastereoselectivity (2.7:1−5.6:1 dr) with high regio- (91:9−>95:5 B/L) and enantiocontrol (89:11−97:3 er). A number of benzyloxyallene derivatives reacted well with amino acid **1a** leading to the desired products (**3af−3ai**) in good yields and regio- and enantioselectivities. Using the bulkier 1,1-dimethylpropyloxyallene and *tert*-butyl-oxyallene, the yields of related products (**3aj−3ak**) slightly decreased, while the enantioselectivity remained high. Gratifyingly, alkoxyallene bearing a chiral alkoxyl group on the side chain underwent this allylation reaction well to give the corresponding product **3al** in 83% yield with >19:1 dr for the *syn*-isomer using ***ent*-L9** as the ligand. To investigate the matched/mismatched effect of this transformation, **L9** was used as a ligand for this hydroaminoalkylation reaction, and reversed diastereoselectivities were observed, indicating that the stereochemical induction in this cross-coupling reaction was mostly controlled by the chiral ligand rather than the chiral substrate. To be noted, 1,1′-disubstituted alkoxylallene (**2m**) was also applicable in this transformation.

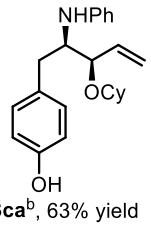
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Table 2. Scope with respect to the amino acids[c]

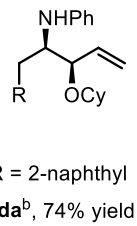




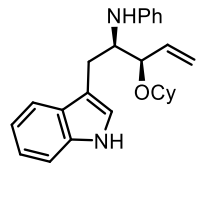
3ba, 70% yield
3.8:1 dr, 87:13 B/L
96:4/89:11 er



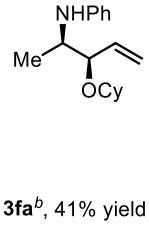
3ca[b], 63% yield
5:1 dr, 91:9 B/L
96:4/90:10 er



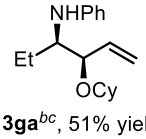
3da[b], 74% yield
4.2:1 dr, 87:13 B/L
96:4/87:13 er



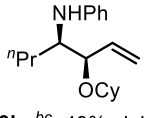
3ea, 59% yield
3.6:1 dr, 93:7 B/L
96:4 er/92:8 er



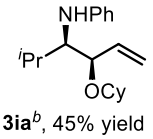
3fa[b], 41% yield
3.1:1 dr, 90:10 B/L
94:6/86:14 er



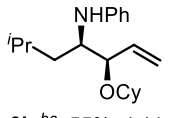
3ga[bc], 51% yield
3.6:1 dr, 87:13 B/L
96:4/88:12 er



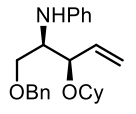
3ha[bc], 49% yield
4:1 dr, 87:13 B/L
96:4/89:11 er



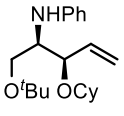
3ia[b], 45% yield
4:1 dr, 88:12 B/L
95:5/74:26 er



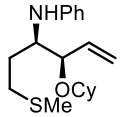
3ja[bc], 55% yield
4.3:1 dr, 88:12 B/L
97:3/90:10 er



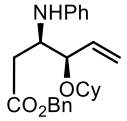
3ka[b], 54% yield
3.2:1 dr, 89:11 B/L
94:6/81:19 er



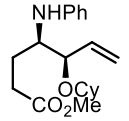
3la, 63% yield
4.5:1 dr, 87:13 B/L
95:5/81:19 er



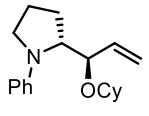
3ma, 49% yield
3.5:1 dr, 88:12 B/L
96:4/87:13 er



3na[b], 58% yield
5.3:1 dr, 88:12 B/L
95:5/84:16 er



3oa[b], 53% yield
3.8:1 dr, 89:11 B/L
96:4/85:15 er



3pa, 44% yield
4.1:1 dr, 84:16 B/L
93:7/85:15 er

[a]Reactions were carried out with reaction conditions in Table 1, entry 2, unless otherwise noted. Isolated yields are presented. Diastereomeric ratio (dr) was determined by ^{1}H NMR spectroscopy of the crude mixture. Er was determined by chiral HPLC analysis. [b][**PC1**] (1 mol %) was used. [c]Er was determined after derivatization of corresponding compounds (see the Supporting Information).[19]

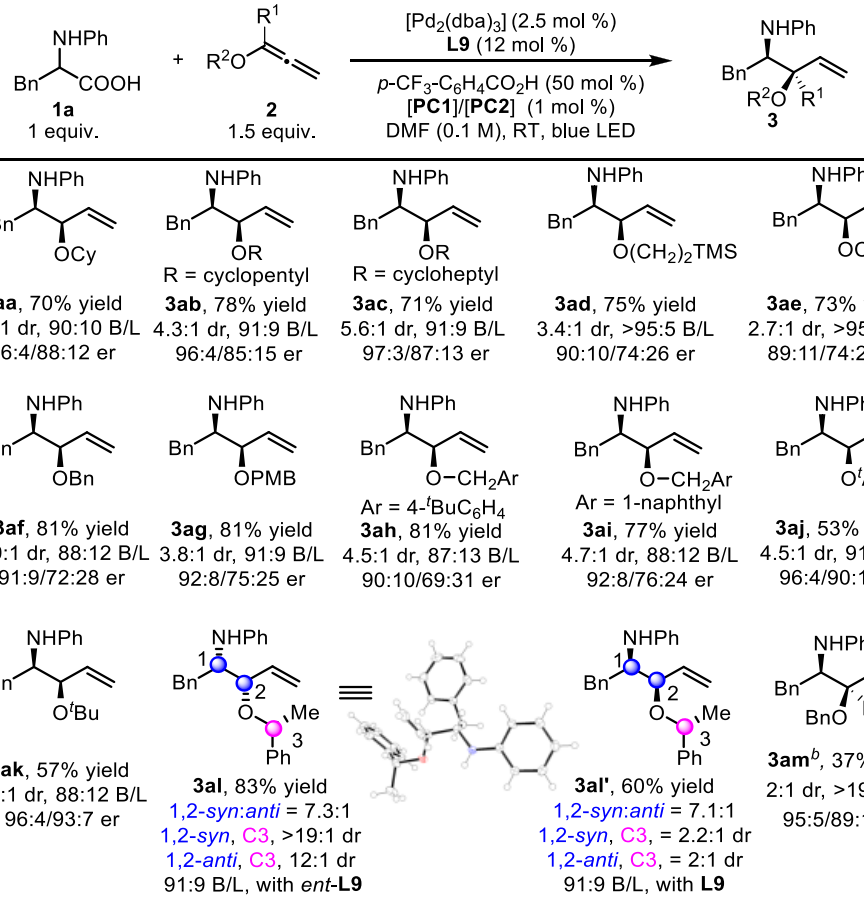
According to previous observations on palladium-catalyzed addition of pronucleophiles into allenes,[20] by switching the ligand to *rac*-BINAP, to our delight, the alkyl-substituted terminal allenes coupled well with amino acids, leading to an array of homoallylic amines (**4a**−**4j**, **4l**−**4t**) with exclusive linear- and *E*/*Z* selectivities (Table 4). The terminal allenes containing various functional groups, such as OH, OTBS, phthalimide, and alkene, were well tolerated in this transformation. The phenyl allene coupled well in this transformation, although the related product (**4k**) could only be obtained in a modest L/B ratio. Satisfyingly, the reactions with PMP-protected amino acids performed well, i.e., delivered the desired adducts (**4s**−**4t**) with up to 90% yields. The generality of this transformation offers a direct and efficient strategy to prepare homoallylic amines.

Furthermore, the practical utility of this methodology was highlighted by several postcatalysis transformations of the furnished products (Scheme 2). Compound **3al** was subjected to acidic conditions to generate the chiral vicinal amino alcohol **5a** in 85% yield with 99:1 er. Hydrogenation of **3ag** with Pd/C under H_2 (1 atm) at ambient temperature followed by subsequent treatment with trifluoroacetic acid (TFA) furnished the secondary alcohol **5b** in a good yield. Hydroboration and consecutive oxidation of **3aa** gave the 1,4-amino alcohol **5c** in 63% yield. Rhodium-catalyzed hydroformylation of **3aa** could form a six-membered cyclized enamine, which could be hydrogenated in situ affording 2,3-disubstituted piperidine (**5d**) in 60% yield over two steps.

To gain some insights into the mechanism, a series of experiments were carried out (see the Supporting Information). The addition of a radical scavenger (TEMPO) could inhibit the reaction completely, which indicated the prevalence of a free-radical pathway. Furthermore, a radical trap experiment was conducted with the addition of ethyl acrylate, and the α-alkylated product was formed through an α-amino alkyl radical via a Giese-type reaction. The application of *N*-phenyl amino acid (L-**1a**) or (D-1a) derived from enantiomerically pure phenyl alanine in this transformation leads to the same regio- and stereoselectivities. The coupling reaction between **1a** and cyclohexyl allene under the linear-selective reaction conditions produced a dimer compound from **1a**. Those observations also support the formation of an intermediate α-amino alkyl radical. Stern−Volmer studies of *N*-phenyl amino acid (**1a**) with photocatalyst [**PC1**] revealed that a simple *N*-phenyl amino acid

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Table 3. Scope with respect to Alkoxyallenes[b]



[a]Reactions were carried out with reaction conditions in Table 1, entry 2. [b][**PC1**] (1 mol %) was used.

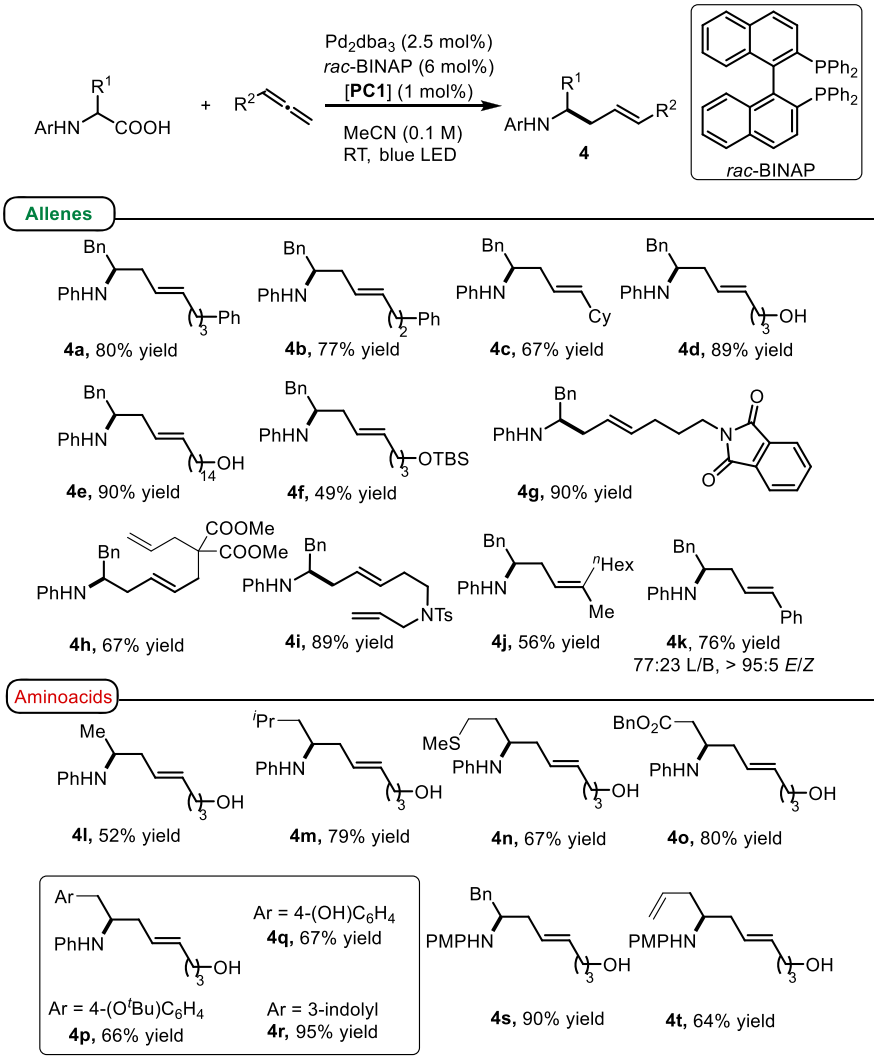
(**1a**) was not able to quench the photoexcited Ir^{III} complex, while basifying **1a** with Cs_2CO_3 could effectively undergo a reductive quenching process. Although there was no base additive in the reaction system, the formation of π-allyl palladium intermediate from allenes will promote the deprotonation of amino acids, which also hints that a π-allyl palladium intermediate is formed prior to the single-electron decarboxylation of amino acids. Furthermore, the low quantum yield suggested that a radical-chain propagation mechanism was not involved in this reaction.[18] In addition, the importance of irradiation throughout the reaction was further verified by a light on/off experiment. When monitoring the reaction by ^1H NMR within the first 1 h, a fast addition of the carboxylic acid to allene forming the ester **5aa** was observed. This suggested that the oxidative decarboxylation occurs after this ester formation.

Based on these mechanistic studies and previous reports, a potential mechanism for this decarboxylative hydroaminoalkylation is proposed (Scheme 3). This transformation starts from the oxidative addition of palladium catalyst with amino acid generating the palladium hydride species **A**. Then, migratory insertion of the intermediate **A** to the allene generates the π-allyl palladium species **B**, which then undergoes competing pathways between a fast reductive elimination leading to an ester intermediate (**5aa**) and a relatively slow decarboxylation via a reductive quenching process of the blue-light-excited Ir(III)* complex. A recent report from the Tunge group[21] suggested that in this process, a single electron transfer from the carboxylate to Pd in intermediate **C** can facilitate the decarboxylation process and the newly formed carbon radical then rebounds with the π-allyl-Pd species, resulting in the formation of intermediate **D**. The π-allyl palladium(III) species **D** undergoes reductive elimination forging the formation of a new C−C bond and a Pd(I) species. The catalytic cycle finishes with the regeneration of Pd(0) via the reduction of Pd(I) by the Ir(II) complex.

To conclude, we developed an asymmetric dual photoredox/palladium-catalyzed decarboxylative hydroaminoalkylation of alkoxylallenes to give synthetically useful protected vinyl 1,2-amino alcohol products with high degrees of regio- and enantioselectivities. A variety of common protecting groups as well as a broad range of protected amino acids bearing different functional groups were applicable. Mechanistic experiments demonstrated that oxidative decarboxylation might occur after the ester formation of allenes with amino acids, which is also in accordance with the observation of specific chemoselectivity in the reaction with tyrosine. This transformation enables a different retrosynthetic strategy for accessing enantioenriched 1,2-amino alcohols, complementing the traditional two-electron polarity reversal approaches. With alkyl- and aryl-substituted terminal allenes as the substrates, linear-selective decarboxylative hydroaminoalkylation is realized in high levels of regio- and E/Z selectivities with a Pd/*rac*-BINAP catalytic system. To the best of our knowledge, this transformation represents a rare example of asymmetric hydroaminoalkylation of allenes. Further studies to apply this strategy to synthesize other important bioactive compounds and potential drug candidates are currently ongoing in our laboratory.

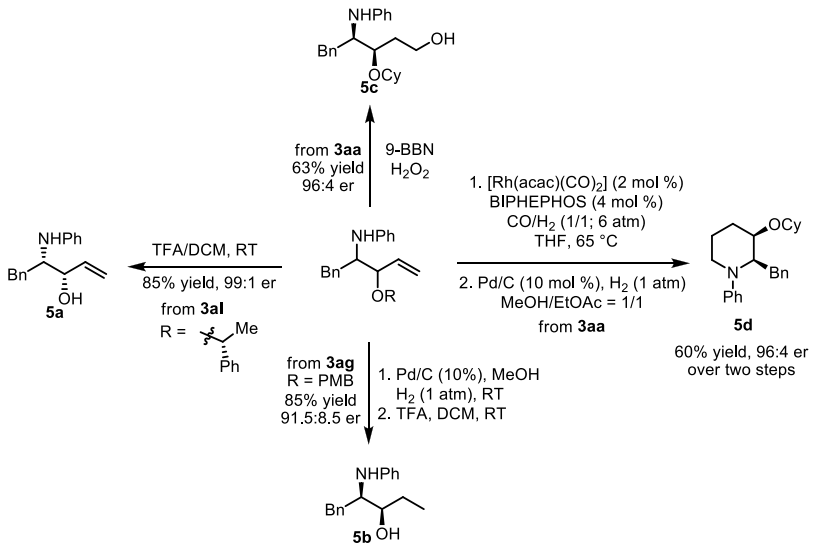
https://dx.doi.org/10.1021/acscatal.1c00153
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Table 4. Scope of Decarboxylative Addition of Amino Acids to Terminal Allenes toward Achiral Linear Homoallylic Amines



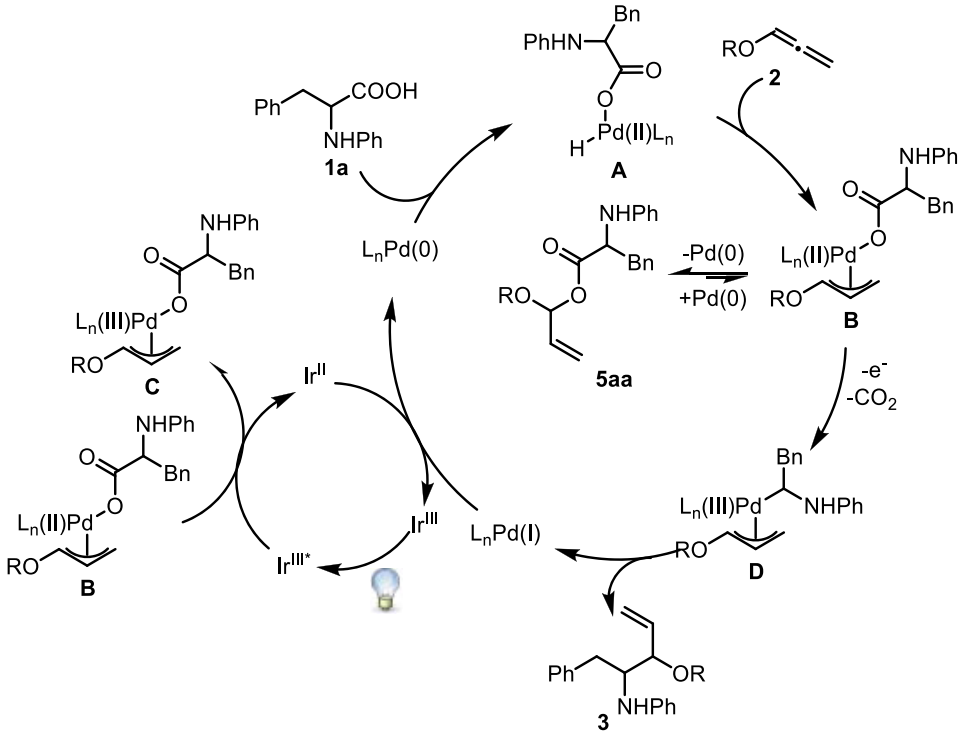
[a]Reaction conditions (unless otherwise specified): **1** (0.2 mmol), **2** (1.2 equiv), [$Pd_2(dba)_3$] (2.5 mol %), *rac*-BINAP (6 mol %), [**PC1**] (1 mol %), MeCN (0.1 M), 16 h, rt, blue LED. Isolated yields are presented. Unless otherwise specified, all of the reactions were performed with >95:5 L/B and ≫95:5 *E*/*Z*.

Scheme 2. Synthetic Application of the Allylic Adducts



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Scheme 3. Proposal Mechanism



■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.1c00153.

Experimental procedures, characterization of all new compounds, and X-ray data for compounds (2*S*,3*S*)-**3al** (CCDC 1999447) (PDF)

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Author Contributions

All authors have given approval to the final version of the manuscript.

Notes

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